Impact of the Pumarejo Bridge on transport costs in the manufacturing industry of the Colombian Caribbean*

Impacto del Puente Pumarejo en los costos de transporte en la industria manufacturera del Caribe Colombiano.

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Abstract

This paper estimates the impact of Colombia's New Pumarejo Bridge on manufacturing firms' transportation cost intensity in the Caribbean region. Using firm-level data from the Annual Manufacturing Survey and a two-way fixed effects difference-in-differences design, with entropy balancing as a robustness check, we find a statistically significant reduction ranging from 8.1% to 22.1% across specifications. Given that transportation dominates Colombia's logistics cost structure, standing at 15.6% of firm sales, above OECD benchmarks, this constitutes a first-order competitiveness gain that also narrows the logistics cost disadvantage historically faced by Caribbean manufacturers relative to more connected regions.

Keywords: Transportation costs; manufacturing sector; difference-in-differences.

JEL Classification: R42. C23. H54. L60. O18.

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1. Introduction

National physical capital, in particular transport infrastructure, is an essential pillar for economic development, regional integration, and the competitiveness of territories. In Colombia, the Caribbean region has historically faced structural limitations in road connectivity and logistics that have affected the efficiency of domestic trade and the competitiveness of its firms (Aguilera-Díaz et al., 2013). In this context, the construction of the *New Pumarejo Bridge*, inaugurated on 20 December 2019, marks a turning point for Caribbean firms and, likewise, one of the country's most significant road infrastructure interventions of the last decade. This project, with an investment of roughly 170 million dollars (INVIAS, 2015), replaced the old structure known as the *Laureano Gómez Bridge*, or *Old Pumarejo Bridge*, built in 1974, which represented a bottleneck for river and land transport in the Caribbean corridor. The project aimed to improve communication between Barranquilla and Sitionuevo, two strategic points in the economic hub of the northern part of Colombia.

It is worth noting that the New Pumarejo Bridge was not the only major infrastructure intervention in the region during this period. As part of the Fourth Generation (4G) concession program, the Cartagena–Barranquilla corridor and its Circunvalar de la Prosperidad component were developed concurrently: the Malambo–Galapa segment (17 km) became operational in December 2018, and the remaining sections of the Circunvalar (36.7 km in total) were progressively delivered through September 2019, with further segments of the broader corridor completed in subsequent years. Since these projects are concentrated within the metropolitan area of Barranquilla and the Bolívar-Atlántico corridor, their timing and geographic overlap with the Pumarejo Bridge must be taken into account when interpreting the estimated treatment effect, as discussed in the Methodology and Results sections.

The new bridge, with a length of 3.2 kilometers and a dual carriageway design with six lanes, was intended to become an emblematic engineering project and, at the same time, a public policy commitment aimed at optimizing mobility, reducing transit times and strengthening the integration of the Colombian Caribbean ports with the interior of the country (Rodríguez Sánchez and Rojas Barrera, 2018). Its relevance transcends the local dimension, as the bridge is a key link in the national logistics corridor, connected to the Ruta del Sol, the Vía al Mar, and the ports of Santa Marta, Barranquilla, and Cartagena. Furthermore, from a socio-economic point of view, the project directly benefits more than 2.1 million inhabitants of the Barranquilla metropolitan area and nearly 8.2 million people in the departments of Atlántico, Magdalena, Bolívar, Cesar, and La Guajira (DANE, 2025c).

The assessment of the impact of the *Pumarejo Bridge* on transport costs in the manufacturing sector of the Colombian Caribbean is, therefore, an exercise in public policy evaluation focused on the impact component. Its approach is in line with the need to measure concrete results on economic variables derived from infrastructure investments, which is consistent with the international literature showing how improvements in connectivity reduce logistics costs, boost total factor productivity, and generate positive externalities on manufacturing activity (Chen et al., 2023; Xu and Feng, 2022).

Despite this growing body of evidence, relatively little is known about whether transport infrastructure effectively reduces firms' transportation expenditures at the microeconomic level, particularly in developing countries (Holl, 2016). Throughout this paper, we use the term transportation costs to refer specifically to firms' direct expenditures on moving inputs and outputs, while logistics costs are reserved for the broader concept used in the literature we review, which also includes components such as warehousing, inventory holding, and order processing that lie outside the scope of what we measure. Most existing studies focus on aggregate outcomes, while the direct cost channel, through which infrastructure is expected to operate, remains underexplored. In the Colombian context, the literature has primarily examined macroeconomic and regional effects of infrastructure investment, leaving firm-level transportation costs largely unaddressed (Peñaloza Blanco, 2017; Sánchez, 1993). This gap raises an important question: *Does major transport infrastructure effectively reduce transportation costs for firms located in the affected region?*

This paper contributes to the literature by providing firm-level causal evidence on the direct impact of a major transport infrastructure project on transportation costs. Using detailed manufacturing survey data and a quasi-experimental difference-in-differences design, we isolate the effect of the New Pumarejo Bridge on firms' transportation expenditures. By focusing on transportation costs as the dependent variable, the analysis identifies the mechanism through which infrastructure improvements may enhance competitiveness, rather than inferring such effects indirectly through broader outcomes.

Methodologically, the research adopts a quasi-experimental design based on the difference-in-differences technique, widely used to estimate causal effects in *ex-post* evaluations of public policies when randomized experiments are not possible (Ateeq-ur-Rehman, 2017). This method compares the evolution of transport costs for manufacturing firms in the departments of Atlántico, Magdalena, La Guajira, and Bolívar, as a treatment group, with those of other regions of the country, before and after the bridge became operational, i.e., at the end of 2019.

These departments play a strategic role in the Colombian economy due to their concentration of manufacturing activity, port infrastructure, and logistical connections to both domestic and international markets. The three most important ports of the country are located in this region, in the cities of Barranquilla, Cartagena, and Santa Marta (Molina et al., 2022). In 2024, approximately three-quarters of all the country's port cargo was moved in this area (MinTransporte, 2025). Additionally, in terms of production, Bolívar and Atlántico are among the most significant departments, ranking sixth and seventh, respectively, in their contribution to GDP. Together, these four departments account for 10% of the national product (DANE, 2025a).

The methodological approach is based on the assumption of parallel trends (Lee, 2016), according to which, in the absence of the bridge, the transport costs of firms in the Caribbean would have followed a similar trajectory to those of firms in other regions. Thus, the interaction between the treatment variable and the post-intervention time variable allows us to isolate the average effect attributable to the infrastructure improvement, which, combined with the inclusion of additional controls, improves the accuracy of the estimate, reduces bias, and allows us to distinguish the direct impact of the bridge's construction on the target variable.

The paper makes three main contributions to the literature. First, it provides micro-level causal evidence on the direct impact of transport infrastructure on firm-level transportation costs. While a growing body of quasi-experimental literature has established that transport infrastructure investments affect aggregate outcomes such as trade flows, regional growth, job creation, and labor income (Chatterjee et al., 2025; Donaldson, 2018; Donaldson and Hornbeck, 2016; Gertler et al., 2024), these studies typically infer cost reductions indirectly through such downstream outcomes rather than measuring the transportation cost channel directly. By constructing a firm-level measure of transportation cost intensity from manufacturing survey data, this paper provides direct evidence on this specific mechanism, rather than inferring it indirectly through trade or income effects.

Second, it exploits a clearly defined quasi-experimental setting based on the opening of a major infrastructure project, in which the entire treatment group is exposed to a single, precisely dated shock rather than a staggered rollout across regions or time periods. This uniform treatment timing avoids the identification complications associated with staggered difference-in-differences designs that have received increasing attention in the recent econometric literature (Roth et al., 2023) and provides an unusually sharp source of identifying variation for a developing-country infrastructure setting.

Third, it contributes to the Colombian literature, which has so far examined the effects of transport infrastructure primarily at the macroeconomic and regional level (Peñaloza Blanco, 2017; Sánchez, 1993), by providing what is, to our knowledge, the first firm-level causal estimate of the direct cost channel through which infrastructure is theorized to affect domestic competitiveness. In doing so, it shifts the focus of the domestic literature from aggregate outcomes to the specific cost mechanism, offering a concrete magnitude that can inform the evaluation of similar infrastructure investments in the region.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data and empirical strategy. Section 4 presents and discusses the results. Section 5 concludes with policy implications.

2. Literature and Empirical Framework

Physical transportation infrastructure plays a fundamental role in economic activity, as its primary objective is to connect territories in order to enhance productive linkages and foster both intraregional and interregional trade (Skorobogatova and Kuzmina-Merlino, 2017). Beyond this connective function, infrastructure can be conceptualized as a productive input that generates positive externalities by reducing transaction costs, increasing the marginal productivity of private capital, and strengthening regional competitiveness, as originally emphasized in the growth literature (Aschauer, 1989). In this sense, infrastructure investments do not merely facilitate movement but actively shape firms' cost structures and production possibilities.

More directly, transportation infrastructure exerts a significant impact on firms by affecting transport costs, travel times, logistical components, and the physical constraints that limit effective access to new markets (Holl and Mariotti, 2018). Improvements in road infrastructure reduce delays, optimize supply chain efficiency, and enhance firms' operational performance, particularly in manufacturing sectors where transportation represents a substantial share of total costs (Liu et al., 2020). In fact, road infrastructure quality is a significant driver of supply chain cost reduction among manufacturing firms, with road safety acting as a complementary mechanism that reinforces this effect (Adu and Dorasamy, 2024). These effects are particularly relevant in regions characterized by high dependence on terrestrial and fluvial transport systems, where logistical inefficiencies tend to be more pronounced due to the non-existent benefits of agglomeration (Lee, 2021).

In classical models of international trade, transportation costs are typically represented through the concept of iceberg costs. Under this framework, when a product is exported, a fraction of it does not reach its destination, i.e., part of the shipment "melts away" in transit (Samuelson, 1954). Within this setting, higher transportation costs reduce the effective quantity of goods delivered to the destination market and increase the relative prices faced by consumers and producers. Conversely, a reduction in such costs expands the effective size of markets, intensifies competition, and alters the allocation of resources across regions.

The New Economic Geography further develops this insight by emphasizing that transportation costs, market size, and economies of scale jointly determine whether production activity concentrates in clusters or becomes spatially dispersed (Krugman, 1991). As transportation costs decline, firms gain access to larger markets, which fosters agglomeration processes and reinforces regions with initial advantages. Infrastructure, therefore, modifies the spatial distribution of economic activity by reducing the friction of distance and redefining specialization patterns across regions (Fujita et al., 2001). According to this theory, reductions in transport costs increase market access but also intensify local competition, disproportionately benefiting sectors with pre-existing comparative advantages and thereby accelerating regional specialization (Pi and Wang, 2025).

More recent contributions extend this framework by incorporating network-based approaches to production. Improvements in connectivity not only reduce direct transportation costs but also reshape the structure of inter-firm relationships and production networks. Enhanced infrastructure facilitates the formation of denser and more efficient supply chains, increases the intensity of input-output linkages, and expands the geographic scope of productive interactions (Bernard et al., 2019). In this context, infrastructure operates as a mechanism that transforms the architecture of economic networks, rather than merely lowering costs in isolation.

Additionally, reductions in transportation and interaction costs contribute to diminishing what has been described as the tyranny of distance, enabling more efficient economic relationships across geographically separated agents (Petersen and Rajan, 2002). Although originally framed in the context of financial intermediation, this argument extends to industrial organization: as spatial frictions decline, firms can access broader markets, diversify suppliers, and strengthen integration within regional and national economies.

In models with heterogeneous firms, reductions in transportation costs generate firm-level reallocation effects: more productive firms expand and gain market share, while less efficient firms face increased competitive pressure (Melitz, 2003). Improved connectivity can also stimulate firm entry in newly accessible markets (Eaton et al., 2011) and promote productivity gains through better access to inputs and demand (Duranton et al., 2014; Melitz and Ottaviano, 2008). These dynamics highlight that infrastructure-induced cost reductions have both direct and indirect effects on industrial performance.

However, before the longer-run, indirect effects described above can fully materialize, the initial impact of infrastructure improvements is reflected in a direct, short-run reduction in transportation costs. Road infrastructure decreases travel times, fuel consumption, maintenance costs, and risks associated with delays, while also lowering coordination costs within supply chains (Minken and Johansen, 2019). We refer to this direct cost reduction as the primary transmission channel in the short run: it is the first-order mechanism through which infrastructure affects firm behavior immediately following an intervention and the entry point from which the longer-run dynamics discussed above are theorized to unfold over time. The present paper focuses specifically on this short-run, direct channel. As discussed in the Results and Discussion section, the estimated reduction in transportation costs is best interpreted as the first stage of a broader, longer-run process of industrial restructuring, rather than as a full account of the infrastructure's general-equilibrium effects.

Recent empirical literature has increasingly focused on identifying the causal effects of transportation infrastructure using quasi-experimental strategies (Donaldson, 2018; Donaldson and Hornbeck, 2016; Kaiser and Barstow, 2022; Lu et al., 2022; Rosik and Wójcik, 2022). More recent contributions continue to expand this evidence base in developing-country settings: Gertler et al. (2024) exploit an instrumental-variables strategy based on Indonesia's road-budgeting process to show that road quality improvements help manufacturers create jobs and raise labor income, while Chatterjee et al. (2025) use the staggered completion of highway segments in India to document how transportation cost reductions reshape labor demand within manufacturing firms and Adu and Dorasamy (2024), using survey data from Ghanaian manufacturers, find that road infrastructure quality is associated with lower perceived supply chain costs.

While this literature confirms that transportation infrastructure affects firms through several distinct channels, direct, administratively measured estimates of the transport-cost-to-output margin itself, as opposed to related outcomes such as employment, wages, or self-reported cost perceptions, remain limited, particularly outside Asian and African contexts.

This gap is particularly relevant in developing economies, where domestic transportation costs often represent a significant share of value added. In these contexts, competitiveness depends not only on international conditions but also on the efficiency of internal logistics systems. The Colombian case illustrates this gap with particular severity: national logistics costs stood at 15.6% of firm sales in 2024, down from 17.9% in 2022 but still well above the 6%–10% range typically observed in OECD economies, and the country ranks 66th out of 139 countries in the World Bank's Logistics Performance Index, having fallen eight positions since 2018 (Departamento Nacional de Planeación, 2024; World Bank, 2023).

Transportation is the single largest component of this cost, accounting for 44.5% of total logistics expenditure in 2024. In the Colombian case, geographical constraints, territorial dispersion, and port concentration contribute to relatively high domestic logistical costs (Sánchez, 1993; Sanchez Polanco et al., 2023). Notably, the Caribbean Central Region (comprising Atlántico, Magdalena, and Bolívar, the departments examined in this study) is identified by Colombia's national planning department as a standout performer in its regional logistics index, reflecting its role as a key logistics corridor connecting the country's main Caribbean ports to its interior. While prior studies have documented the importance of infrastructure for trade and regional development in similarly structured developing economies (Barzin et al., 2018; Chauvet and Ferry, 2021), micro-level causal evidence remains scarce and the scale and singularity of the Pumarejo Bridge intervention offer an unusually clean source of identifying variation relative to the more diffuse, staggered infrastructure rollouts studied elsewhere.

According to this perspective, large-scale infrastructure projects can be interpreted as exogenous shocks that alter both transportation costs and the spatial configuration of economic activity. In particular, improvements in strategic corridors not only reduce logistical bottlenecks but also reconfigure connectivity between regions, facilitate the circulation of goods, and enable deeper integration of production networks. According to the New Economic Geography and production-network literatures (Krugman, 1991; Melitz, 2003), these dynamics are theoretically expected to extend beyond immediate cost savings, eventually influencing the organization of manufacturing activity, supplier relationships, and market expansion. However, such effects typically operate through general-equilibrium channels that unfold over a longer time horizon than the direct cost-reduction effect estimated in this paper. Our empirical strategy is designed to identify the latter, the short-to-medium-run reduction in transport cost intensity, while the broader organizational and network-level effects remain a theoretically motivated expectation rather than a finding directly tested here, and constitute a relevant avenue for future research using complementary outcome measures and a longer post-treatment panel.

This study contributes to the literature not by providing the first firm-level estimate of transport infrastructure effects, but by directly estimating, using administrative panel data and a sharp quasi-experimental shock, the effect of a major infrastructure intervention on the specific transport-cost-to-output margin at the firm level, a channel that, to our knowledge, has not been directly measured in this way for a Latin American economy. In doing so, it complements the existing firm-level evidence on related outcomes (employment, wages, perceived costs) with direct, administratively sourced cost evidence, providing a more precise microeconomic test of the cost-reduction channel that the broader literature has largely inferred indirectly through aggregate trade and regional growth outcomes.

3. Methodology

This research adopts a quantitative, explanatory approach based on a quasi-experimental design. Using data from the Annual Manufacturing Survey (EAM) (DANE, 2025b), which collects information on Colombian manufacturing firms, we construct a firm-level panel covering 2013 to 2023 to estimate the impact of the *New Pumarejo Bridge*.

We use a difference-in-differences model as our empirical strategy. Following Roth et al. (2023), we estimate a two-way fixed effects panel data model (see equation 1). We use robust standard errors with White's correction, since the large variation in firm size can generate heteroskedasticity (Gujarati, 2002).

$$Trans_{it} = \beta_i + \alpha_t + \beta_1 D_i T_t + \beta X_{it} + \gamma Z + u_{it} \tag{1}$$

In the above equation, the dependent variable is the firm's transportation cost intensity, defined as the ratio of total transportation costs (the sum of transportation costs for materials and for products) to total production value, calculated for each firm and each period. We use this ratio, rather than total transportation costs in levels, because transportation costs naturally scale with firm size and production volume; the ratio instead captures the efficiency of transportation spending relative to output, independent of firm size. A decrease in the ratio therefore indicates an improvement in transportation cost efficiency, and vice versa. Regarding the treatment of outliers, we drop observations with a transportation cost-to-production ratio exceeding 1 from the sample, consistent with the maximum ratio value of 0.982 reported in Table 1. Also, we apply the same rule to the export- and import-intensity control variables and we do not apply additional percentile-based winsorization; however, since the dependent variable enters the regression in logarithmic form, the influence of remaining extreme values in the right tail of the distribution is substantially attenuated relative to estimating the model in levels.

Table 1: Descriptive statistics related to transportation costs (In USD)

Variable	*Mean*	*Median*	*Std. Dev.*	*Min. Value*	*Max. Value*
Materials' Transportation Costs	27,391	114	352,841	0	41,592,899
Products' Transportation Costs	112,071	5,602	556,502	0	39,130,161
Total Transportation Costs	139,462	10,706	691,212	0	42,413,811
Production	5,527,477	767,068	36,941,061	3,618	3,092,103,643
Ratio	0.027	0.015	0.043	0.00	0.982

Source: Own elaboration using information from AMS.

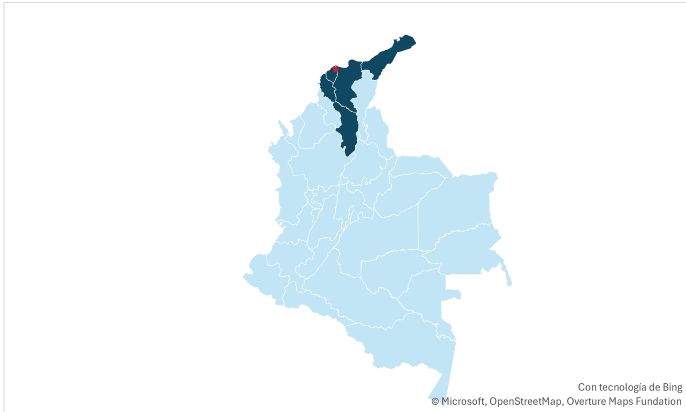
It is worth noting that the data used has a relevant limitation regarding the study of price effects. While the survey records the total monetary value of inputs purchased and outputs sold by each firm, it does not report the corresponding physical quantities. As a result, it is not possible to construct genuine unit values for inputs or outputs, since any ratio built solely from the value variable would confound changes in prices with changes in production or sales volume. For this reason, the present analysis focuses on transportation cost intensity as the outcome of interest, rather than on price effects.

A related limitation concerns the geographic granularity of the dataset, which identifies firms only at the department level and does not report the municipality in which each firm is located. This precludes a robustness check that would exclude firms located specifically in the municipalities directly served by the Circunvalar de la Prosperidad and the broader Cartagena–Barranquilla corridor, i.e., Malambo, Galapa, Puerto Colombia, and Cartagena, which became operational on a similar timeline to the bridge. As a feasible alternative given the available data, the results and discussion section presents a robustness check that restricts the treatment group to firms located in Magdalena, the department whose connectivity gain stems primarily from the bridge itself rather than from the Atlántico-Bolívar metropolitan corridor improvements.

The variable D_i is a dichotomous variable referring to whether the firm is part of the treatment or control group. According to Peñaloza Blanco (2017)'s research, the bridge has a direct influence area of twenty cities, which are located in the departments of La Guajira, Magdalena, Bolívar, and Atlántico (although La Guajira is not represented in the survey, so only firms from the other three departments are included).

This twenty-city area is not a separate unit of analysis from the departments discussed in the Introduction, but rather their urban-economic core: Atlántico, Bolívar, and Magdalena jointly account for approximately 63% of the Caribbean region's GDP and roughly 9.5% of national GDP (DANE, 2025a), with this economic activity heavily concentrated in the metropolitan areas of Barranquilla, Cartagena, and Santa Marta that anchor the bridge's influence area. The treatment group used in our empirical strategy, defined at the department level due to data availability, therefore captures the firms located within and around this urban-economic core. Figure 1 displays the three departments used as the treatment group in the empirical strategy; because the twenty-city influence area is a subset of these departments and firm location is not observed at the municipality level, the department-level boundaries shown in the figure represent the empirical treatment group, which is broader than, but encompasses, the twenty-city zone of direct influence.

Figure 1: Pumarejo Bridge influence area (in departments)



Source: Own elaboration using Microsoft Excel.

In order to construct the treatment group, we consider firms from these departments, while the rest of the country serves as the control group. In the second equation, the variable T_t is a temporal reference, also dichotomous, which takes values of 1 for data from 2020 onwards and 0 for values prior to this year. Thus, β_1 is the quantification of the impact.

In addition, as a control, we add a vector of variables X that may be related to the firm's transportation costs, such as productivity, wages, markup, external sector, market share, capital-labor ratio, technological sectors, and the firm's size.

We measure productivity as the logarithm of the total factor productivity of the firm, following the Wooldridge method (Wooldridge, 2009), using the production factors: capital, labor, materials, energy, and water following the existing theoretical concepts (Baily, 1986; Wei, 2007; Zhang et al., 2017). We measure the external sector with two ratios of exports and imports intensity. Also, we measure wages as the logarithm of the wages per capita of the firm. We use the market share as a proxy for the market power of the firm, and it is measured as the proportion of the firm's production in the total sector production. Additionally, in order to capture the firm's size into the equation, we use two dummies for the medium and large firms, and to account for the different technological sectors, using the categorization of Gómez-Sánchez et al. (2022), we use two other dummies to characterize the medium and high-tech sectors. We further include a vector Z, referring to the different sectors within manufacturing through two-digit CIIU codes. We estimate the model using STATA software (Correia, 2023).

Table 2 presents descriptive statistics for treated and control firms measured in 2019, the year immediately preceding the bridge's inauguration, in order to characterize the two groups prior to treatment without contamination from post-treatment dynamics. While the parallel trends assumption (validated in Appendix 1) concerns the evolution of the outcome variable rather than the levels of firm characteristics, observable differences in baseline covariates such as productivity, firm size, market power, and trade orientation can still bias a simple comparison of means if left unaddressed, and motivate their explicit inclusion as controls in our regression specification.

Table 2: Pre-treatment balance (2019)

Variable	**Control**	**Treated**	**Diff. (C-T)**	**Std. Err.**
Log transport cost intensity	-4.509	-4.325	-0.184*	(0.107)
Log total factor productivity	1.924	1.854	0.070***	(0.026)
Log market share	-7.199	-7.026	-0.172	(0.121)
Log wages per capita	9.358	9.317	0.041	(0.087)
Capital-labor ratio	12.177	12.448	-0.271***	(0.096)
Log markup	0.397	0.311	0.086***	(0.032)
High-tech sector (dummy)	0.159	0.246	-0.087***	(0.025)
Medium-tech sector (dummy)	0.346	0.358	-0.012	(0.032)
Medium-sized firm (dummy)	0.292	0.375	-0.083***	(0.031)
Large firm (dummy)	0.091	0.121	-0.030	(0.020)
Export intensity	0.058	0.104	-0.046***	(0.011)
Import intensity	0.058	0.114	-0.056***	(0.011)
Number of Firms	**4,232**	**4,302**		

Source: Own elaboration using information from AMS. **Note:** *** $p<0.01$, ** $p<0.05$, * $p<0.10$

As shown in Table 2, treated and control firms differ significantly, along 10 of the 14 covariates examined: firms in the treatment group exhibit higher total factor productivity, a lower capital-labor ratio, higher markups, lower representation in high-technology sectors, and a systematically lower propensity to export or import together with a smaller share of medium-sized firms. No significant pre-treatment differences are found in market share, wages, medium-technology classification, or the share of large firms. A small but only marginally significant difference is also present in the outcome variable itself. These baseline imbalances reinforce the relevance of the control vector X and the firm fixed-effects specification used throughout our analysis, rather than relying on a raw, unconditional comparison between the two groups.

To further address these imbalances and assess the robustness of the main results to pre-existing compositional differences between groups, we implement Entropy Balancing (EB) as an auxiliary reweighting strategy (Hainmueller, 2012). This approach avoids specification dependence on a propensity score model, retains all control observations in the reweighted sample, and guarantees exact mean balance rather than approximating it. Using 2019 as the balancing year and the eleven covariates listed in Table 2, we compute entropy weights for the control group and verify that post-balancing differences in all covariate means are negligible (see Appendix 3). We then apply these weights to the TWFE estimator across the same four specifications as in Table 3. The resulting EB+DiD estimates are reported in Appendix 4.

4. Results and discussion

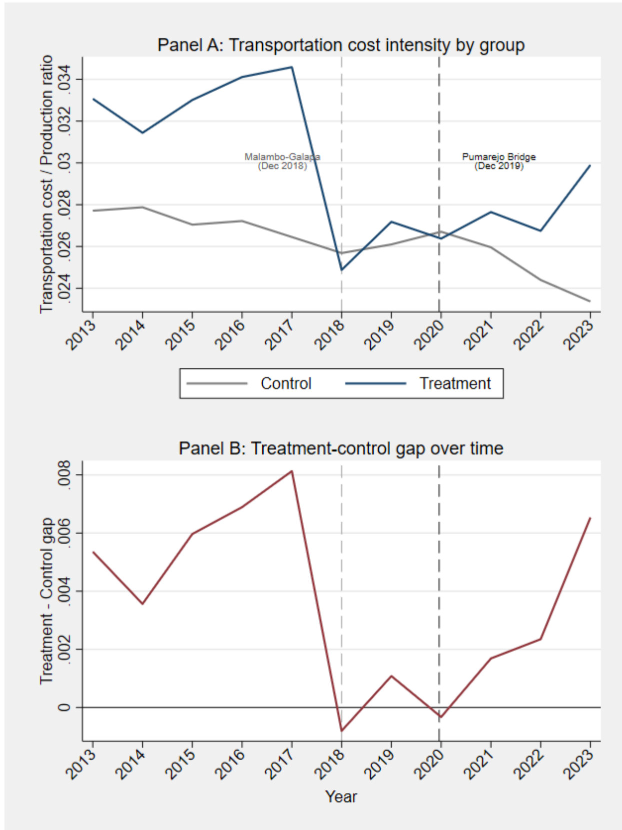
Before presenting the DiD estimates, we validate the parallel trends assumption. To this end, we analyze the evolution of transport costs for manufacturing firms in the Caribbean and other regions during the period prior to the opening of the *Pumarejo Bridge*. We formalize this assumption with an *event study* test (Angrist and Pischke, 2009).

As a first visual check, Figure 2 shows that the assumption appears to hold. Panel A plots the raw transportation cost intensity of Caribbean and other firms over time, while Panel B plots the treatment-control gap directly. On average, the transportation costs of Caribbean firms behave similarly to those of the rest of the country over time.

A visible fluctuation occurs in 2018, when the treatment group's ratio drops sharply before partially recovering in 2019; this coincides with the opening of the Malambo–Galapa segment (December 2018), one of the concurrent 4G corridor projects discussed in the Introduction, rather than reflecting noise or a data anomaly. Both reference points are marked with vertical lines in Panels A and B of Figure 2 below. Importantly, this pre-treatment fluctuation does not threaten our identification strategy because the event-study estimates in Appendix 1 confirm that neither the 2018 nor the 2019 year-treatment interactions are statistically significant, indicating no detectable departure from parallel trends prior to the bridge's inauguration in 2020.

From 2020 onward, Panel B shows that the raw treatment-control gap widens rather than narrows, which may appear, at first glance, to be at odds with the negative and statistically significant difference-in-differences estimates reported in Table 3. This apparent tension is resolved by the baseline imbalances documented in Table 2: treated firms differ systematically from control firms in productivity, markups, trade orientation and other observable characteristics and the unconditional gap in Panel B reflects these compositional differences in addition to any treatment effect. Once firm fixed effects and the control vector X are accounted for, as in columns 2 and 4 of Table 3, the conditional estimate indicates that, holding firm characteristics fixed, treated firms experienced a relative reduction in transport cost intensity that is not directly visible in the unconditional series plotted in Figure 2.

Figure 2: Ratio of transportation costs – Production over time (Panel A: levels by group; Panel B: treatment-control gap)



Source: Own elaboration using Stata.

While the figure provides a useful visual check, formal validation requires an event study. This involves estimating a model in which the coefficients associated with the years prior to treatment must not be statistically different from zero, supporting the validity of the parallel trends assumption.

As can be seen in Appendix 1, there is insufficient statistical evidence to reject the parallel trends hypothesis, since none of the coefficients are statistically significant. Given these results, we consider the impact analysis using the difference-in-differences method robust. Therefore, Table 3 reports the TWFE estimates under alternative combinations of controls and fixed effects.

Table 3: DiD regression results

$Trans_{it}$	(1)	(2)	(3)	(4)
$D_i * T_t$	-0.250***	-0.091*	-0.230***	-0.085*
	(0.066)	(0.050)	(0.048)	(0.050)
TFP_{it}			-1.005***	-0.880***
			(0.059)	(0.060)
$MShare_{it}$			-0.052***	-0.087***
			(0.012)	(0.013)
Med_{it}			0.072**	0.115***
			(0.029)	(0.029)
$Large_{it}$			0.080	0.175***
			(0.054)	(0.054)
Exp_{it}			0.115***	0.130***
			(0.021)	(0.020)
Imp_{it}			0.146***	0.138***
			(0.022)	(0.022)
$Wage_{it}$			0.023***	0.030***
			(0.007)	(0.007)
KL_{it}			-0.017	-0.029*
			(0.018)	(0.018)
$Mkup_{it}$			0.526***	0.447***
			(0.049)	(0.049)
$HighT_{it}$			0.114	0.079
			(0.082)	(0.082)
$MedT_{it}$			-0.038	-0.021
			(0.087)	(0.087)
Constant	-4.487***	-4.482***	-3.236***	-3.634***
	(0.001)	(0.005)	(0.218)	(0.217)
Year FE	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Department Controls	No	Yes	No	Yes
No. of Obs.	53,006	52,093	52,093	52,093
R^2	0.001	0.623	0.628	0.631
R^2 Adj.	-0.185	0.560	0.566	0.570

Note: * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Given that the Pumarejo Bridge was inaugurated alongside other 4G infrastructure projects within the same treatment departments such as the Malambo–Galapa segment (December 2018) and the Circunvalar de la Prosperidad (completed in stages through 2019 and beyond), we recommend interpreting the coefficient in Table 3 with caution, as it may capture the combined effect of the broader 2018–2020 Caribbean corridor infrastructure upgrade rather than the isolated effect of the bridge alone. As a robustness check, and given that firm location is only observed at the department level in the EAM data, we re-estimate the model restricting the treatment group to firms in Magdalena, whose access gains stem primarily from the bridge itself, and excluding Atlántico and Bolívar entirely, since these departments host the concurrent Circunvalar/Cartagena–Barranquilla corridor. As shown in Appendix 2, the estimated coefficient remains negative and statistically significant across all specifications, with a magnitude that is, if anything, larger than that obtained for the full treatment group. This result indicates that the estimated reduction in transport cost intensity is not an artifact of the concurrent 4G corridor projects in Atlántico and Bolívar, and lends additional credibility to interpreting the bridge itself as a primary driver of the observed effect.

As a second robustness check, we address directly the pre-treatment covariate imbalances documented in Table 2 through the Entropy Balancing + DiD estimator described in the Methodology section. After reweighting, all eleven pre-treatment covariates achieve exact mean balance between treated and EB-weighted control firms, as shown in Appendix 3. The EB+DiD results, reported in Appendix 4, are fully consistent with the unweighted baseline: across all four specifications, the estimated coefficient on the DiD interaction remains negative and statistically significant, with similar standard errors. The preferred specification yields a reduction of 10.0% compared to 8.1% in the unweighted TWFE. The EB-weighted estimate is thus slightly larger in magnitude and more precisely estimated, with the level of statistical significance strengthened from 10% to 5%.

This convergence confirms that the main result is not an artifact of compositional differences between treated and control firms, and that the pre-existing imbalances in productivity, capital intensity, markup, and trade orientation documented in Table 2 do not materially bias the TWFE estimates reported in Table 3.

In terms of magnitude, since the dependent variable is expressed in logarithmic form, we interpret the coefficients in Table 3 through the semi-log transformation for dummy variables, $100 \times (e^{\beta} - 1)$ (Halvorsen and Palmquist, 1980), rather than as direct percentage points. This yields an estimated reduction in transport cost intensity ranging from approximately 8.1% (column 4, our preferred specification with full controls and fixed effects) to 22.1% (column 1). Given that the mean transport cost intensity in our sample is 2.7% of production value (Table 1), this corresponds to a fall to approximately 2.5% in the most conservative specification, and to approximately 2.1% in the least conservative one. This magnitude is broadly consistent with recent evidence from comparable developing-country contexts: the World Bank's *Shrinking Economic Distance* report estimates that reducing shipment distance by 100 kilometers lowers transport costs by approximately 20% in low- and middle-income countries (Dappe et al., 2024), a figure closely aligned with our upper-bound estimates. Also, our results are consistent with the theoretical framework that conceptualizes infrastructure as a productive input that reduces transaction costs and enhances firm efficiency (Aschauer,

1989) and with recent empirical findings in the literature.

Beyond statistical significance, the magnitude of the estimated effect is economically meaningful. Transportation expenditures represent a non-negligible share of total operating costs for manufacturing firms in our sample. This reduction, therefore, implies a substantial improvement in firms' cost structure, potentially affecting pricing decisions, margins, and competitiveness. In industries characterized by relatively tight profit margins, such a decline in transportation costs can translate into economically relevant gains in operational efficiency.

It is important to note that this reduction does not simply reflect a secular downward trend in transportation costs. As the event study in Appendix 1 confirms, pre-intervention trends are statistically indistinguishable between treated and control firms, and the decline observed after the opening of the bridge is significantly larger than the variation observed during the pre-treatment period. Moreover, the control group does not exhibit a comparable reduction in costs over the same timeframe. Taken together, the magnitude and timing of the estimated effect suggest that the bridge generated a discrete and economically significant reduction in transportation costs for exposed firms. The stability of the coefficient across specifications further reinforces this interpretation.

From a firm-level perspective, the magnitude of the effect is comparable to estimates found in studies such as Holl (2016), who documents significant productivity gains in manufacturing firms following highway expansions, and Liu et al. (2020), who find that improved transport infrastructure generates measurable reductions in operational inefficiencies. In this sense, the results confirm that reductions in transportation costs are not merely a secondary outcome, but rather a primary mechanism through which infrastructure affects firm performance.

Importantly, the findings provide direct empirical support for the argument that infrastructure reduces logistical frictions by lowering travel times, coordination costs, and uncertainty in supply chains (Minken and Johansen, 2019). Unlike some of the existing literature, like Donaldson (2018), which infers cost reductions indirectly through trade or income effects, this study explicitly captures the cost channel at the firm level. This distinction is non-trivial, as it provides clearer identification of the mechanism linking infrastructure to competitiveness.

Through the New Economic Geography framework, the observed decline in transportation costs suggests an effective reduction in economic distance, which likely enhances firms' access to both input and output markets (Fujita et al., 2001). This mechanism is further supported by evidence from Duranton et al. (2014), where improved road infrastructure significantly increases trade flows by lowering internal trade costs.

Beyond cost reductions, the results are also consistent with the literature on production networks. Bernard et al. (2019) demonstrate that improved connectivity strengthens inter-firm linkages and increases the efficiency of production networks. In the case of the Colombian Caribbean, the Pumarejo Bridge likely reduced bottlenecks in a key logistics corridor, facilitating smoother flows between ports and inland markets. This suggests deeper structural changes in supply chain organization may accompany the estimated cost reduction, even though we do not directly measure these in the present study.

A particularly relevant dimension of the findings is their spatial heterogeneity. The stronger impact observed in the Caribbean region is consistent with the idea that infrastructure investments yield larger marginal benefits in peripheral or poorly connected regions, in line with the evidence presented by Holl and Mariotti (2018), who show that firms located in areas with initially low accessibility experience greater gains from improvements in transport infrastructure.

This interpretation is especially pertinent in the Colombian case, where geographical fragmentation and high domestic transport costs have historically constrained regional competitiveness (Sánchez, 1993). The Caribbean region, despite its strategic port infrastructure, has faced persistent bottlenecks in inland connectivity. The Pumarejo Bridge, by alleviating one of these critical constraints, appears to have reduced the systemic inefficiencies embedded in the regional logistics network.

However, it is important to note that the magnitude of the estimated effect, while economically significant, does not fully capture the long-term general equilibrium impacts of the infrastructure intervention. Previous studies suggest that reductions in trade costs can trigger firm reallocation dynamics, where more productive firms expand and less efficient firms exit (Melitz, 2003). Therefore, the observed reduction in transportation costs may represent only the first stage of a broader process of industrial restructuring.

5. Conclusions and recommendations

This research estimates the impact of the construction of the *Pumarejo Bridge* in terms of transport costs for manufacturing firms. The results show that, in effect, manufacturing firms in the Colombian Caribbean experienced a significant reduction in transport cost intensity following the bridge's inauguration, compared to those in the rest of the country.

We attribute this reduction in costs to several mechanisms. Specifically, improved connectivity between Caribbean ports and the interior of the country, reduced travel times, and less vehicle wear and tear, which translates into significant transportation cost savings. In addition, the bridge acts as a strategic point within the national transport corridor, strengthening the link between the major cities of the Colombian Caribbean.

From a public policy perspective, the results highlight the importance of prioritizing investments in logistics infrastructure with demonstrably high economic impact. The estimated reduction in transport cost intensity ranges from 8.1% in the preferred specification to 22.1% in the most conservative one, which for the average firm in the sample, whose transport-to-production ratio stands at 2.7%, implies a fall ranging from approximately 2.5% under the preferred specification to 2.1% under the most optimistic estimate. This magnitude is economically meaningful because transportation is not a marginal cost item for Colombian manufacturers, it is the single largest component of total logistics expenditure in the country, accounting for 44.5% of logistics costs that already stand at 15.6% of firm sales (Departamento Nacional de Planeación, 2024).

A reduction of the magnitude estimated here in the component that dominates the logistics cost structure therefore implies a first-order improvement in the cost competitiveness of exposed firms, with direct implications for pricing margins, input sourcing, and market reach.

The connection to regional gaps operates through a related but distinct mechanism. Colombia's Caribbean region has historically combined strategic port infrastructure with persistent bottlenecks in inland connectivity, a structural contradiction that has constrained the competitiveness of firms located there relative to firms in more internally connected regions of the country (Sánchez, 1993).

The estimated reduction in transport cost intensity for Caribbean manufacturing firms partially closes this gap: it brings the logistics burden of firms in the region closer to that of firms operating in areas with better pre-existing road endowments, and does so through a mechanism that is particularly valuable precisely where baseline connectivity is weakest. This interpretation is consistent with the finding in the Results section that infrastructure investments yield larger marginal gains in peripheral or poorly connected regions (Holl and Mariotti, 2018) and is further supported by the Magdalena-only robustness check, where the estimated coefficient is larger in magnitude than for the full treatment group, as expected if the bridge's effect concentrates in the department whose connectivity improvement derives most directly from the intervention itself.

In this sense, the Pumarejo Bridge demonstrates how targeted infrastructure investments can attenuate the logistics cost disadvantage that characterizes peripheral regions in developing economies, contributing to territorial integration not through a redistribution mechanism but through a direct reduction in the friction costs that structurally limit the productive potential of firms outside the country's most connected core.

Future research could incorporate methodologies such as heterogeneous differences-in-differences or synthetic control models to explore whether the effects of the intervention vary across firms with different predetermined characteristics. Such extensions would require a careful discussion of the identification strategy and the exogeneity of the interacting variables, which lies beyond the scope of the present study. Future research could also extend the analysis to manufacturing employment effects and the location of new industrial plants, following the line proposed by Obregón-Biosca et al. (2014). Additionally, it could complement firm-level evidence with administrative origin–destination cost data from the Ministry of Transportation to assess route-specific cost dynamics associated with the bridge.

We recommend interpreting our regression results with caution, since the literature has criticized the TWFE difference-in-differences methodology for imposing parametric assumptions and for relying on the fulfillment of specific assumptions (de Chaisemartin and D'Haultfœuille, 2020; Imai and Kim, 2021). Nevertheless, given its relative simplicity compared to alternative methodologies, we adopt it as our main approach.

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Appendices

Appendix 1: Event study

Table A1: Event study

$Trans_{it}$	Event Study Coef. / Rob. Err
$(2013 * D)_{it}$	0.015
	(0.098)
$(2014 * D)_{it}$	0.023
	(0.097)
$(2015 * D)_{it}$	0.002
	(0.098)
$(2016 * D)_{it}$	0.133
	(0.098)
$(2017 * D)_{it}$	0.107
	(0.096)
$(2018 * D)_{it}$	0.025
	(0.092)
$(2019 * D)_{it}$	0.001
	(0.100)
$(2020 * D)_{it}$	-0.099
	(0.107)
$(2021 * D)_{it}$	-0.063
	(0.110)
$(2022 * D)_{it}$	-0.022
	(0.105)
$(2023 * D)_{it}$	0.000
	(.)
Constant	-4.484***
	(0.006)
Year FE	Yes
Firm FE	Yes
Department Controls	Yes
No. of Obs.	37,153
R-Squared	0.843
R-Squared Adj.	0.805

Note: * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Appendix 2: Robustness test – Excluding Cartagena-Barranquilla corridor

Table A2: Robustness test – Excluding Cartagena-Barranquilla corridor

$Trans_{it}$	(1)	(2)	(3)	(4)
$D_i * T_t$	-0.714***	-0.567***	-0.663***	-0.544***
	(0.227)	(0.160)	(0.157)	(0.157)
TFP_{it}			-1.020***	-0.889***
			(0.061)	(0.062)
$MShare_{it}$			-0.048***	-0.084***
			(0.013)	(0.013)
Med_{it}			0.079***	0.122***
			(0.030)	(0.030)
$Large_{it}$			0.055	0.158***
			(0.056)	(0.056)
Exp_{it}			0.115***	0.130***
			(0.021)	(0.021)
Imp_{it}			0.143***	0.136***
			(0.023)	(0.023)
$Wage_{it}$			0.024***	0.030***
			(0.007)	(0.007)
KL_{it}			-0.022	-0.035*
			(0.019)	(0.018)
$Mkup_{it}$			0.544***	0.461***
			(0.051)	(0.051)
$HighT_{it}$			0.122	0.087
			(0.084)	(0.084)
$MedT_{it}$			-0.055	-0.042
			(0.089)	(0.090)
Constant	-4.497***	-4.490***	-3.127***	-3.528***
	(0.000)	(0.005)	(0.225)	(0.224)
Year FE	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Department Controls	No	Yes	No	Yes
No. of Obs.	50,546	49,695	49,695	49,695
R^2	0.001	0.621	0.626	0.630
R^2 Adj.	0.001	0.559	0.564	0.568

Note: * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$

Appendix 3: Pre and post EB covariate balance (2019)

Table A3: Pre and post EB covariate balance (2019)

Variable	Treated	Control (Unweighted)	Control (EB-Weighted)
TFP_{it}	1.854	1.924	1.854
$MShare_{it}$	-7.026	-7.199	-7.027
$Wage_{it}$	9.317	9.358	9.317
KL_{it}	12.447	12.177	12.447
$Mkup_{it}$	0.311	0.397	0.311
$HighT_{it}$	0.246	0.159	0.246
$MedT_{it}$	0.358	0.1346	0.358
Med_{it}	0.375	0.292	0.375
$Large_{it}$	0.121	0.091	0.121
Exp_{it}	0.104	0.058	0.104
Imp_{it}	0.114	0.058	0.114
N	232	4,302	4,302

Note: Entropy weights computed using `ebalance` in Stata, targeting first-moment balance (means) with analytic weights assigned to control units. Treated units receive weight 1.

Appendix 4: EB+DiD regression results

Table A4: EB+DiD regression results

$Trans_{it}$	(1)	(2)	(3)	(4)
$D_i * T_t$	-0.219***	-0.105**	-0.210***	-0.105**
	(0.048)	(0.050)	(0.049)	(0.050)
TFP_{it}			-0.929***	-0.837***
			(0.130)	(0.132)
$MShare_{it}$			-0.069***	-0.089***
			(0.025)	(0.026)
Med_{it}			0.006	0.033
			(0.062)	(0.063)
$Large_{it}$			0.245**	0.299***
			(0.105)	(0.105)
Exp_{it}			0.103**	0.106**
			(0.044)	(0.044)
Imp_{it}			0.149***	0.141***
			(0.046)	(0.045)
$Wage_{it}$			0.025	0.031
			(0.021)	(0.021)
KL_{it}			-0.007	-0.011
			(0.036)	(0.036)
$Mkup_{it}$			0.458***	0.382***
			(0.105)	(0.106)
$HighT_{it}$			0.121	0.080
			(0.232)	(0.204)
$MedT_{it}$			0.023	-0.004
			(0.232)	(0.211)
Constant	-4.305***	-4.325***	-3.513***	-3.813***
	(0.012)	(0.012)	(0.517)	(0.527)
Year FE	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Department Controls	No	Yes	No	Yes
No. of Obs.	41,146	41,146	41,146	41,146
R^2	0.633	0.636	0.643	0.645
R^2 Adj.	0.585	0.588	0.596	0.599

Note: * $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$.
The reduction in observations relative to Table 3 (from 53,006 to 41,146) reflects the exclusion of control-group firms not observed on 2019 and therefore have no assigned entropy weight. Treated firms active post-2019 that were absent from the 2019 balancing cross-section receive weight 1.